Pandrosion Invariants of Knots:
Zeta Functions, Energy, and the Alexander Polynomial

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Abstract

Applying the Pandrosion quotient framework to the Alexander polynomial $\Delta_K(t)$ of a knot K, we obtain a family of Pandrosion knot invariants: the knot zeta function $\zeta_K(s) = \sum \Delta'(\alpha_k)^{-s}$ (satisfying $\zeta_K(1) = 0$ by Paper 30), the Pandrosion discriminant $D^2(\Delta_K) = \prod \Delta'(\alpha_k)$, and the knot energy polynomial $E_{\Delta_K}(t) = \Delta'(t)^2 - \Delta(t)\Delta''(t)$. The palindromic symmetry $\Delta(t) = t^{-d}\Delta(t^{-1})$ implies that the Alexander roots come in pairs $(\alpha, 1/\alpha)$, forcing the Pandrosion quotients into conjugate pairs with real coefficient sums. The Mahler measure $m(\Delta_K)$—which is conjecturally proportional to the hyperbolic volume of the knot complement (Boyd, 2002)—decomposes per root via $m = \sum \log^+ |\alpha_k|$ (Paper 27): knots with all roots on the unit circle (fibered knots) have vanishing Mahler measure. All identities are verified for seven knots (3_1 through 6_3).

1 Pandrosion identities for the Alexander polynomial

Let $\Delta_K(t) = a_d t^d + \cdots + a_0$ be the Alexander polynomial of a knot K, treated as an ordinary polynomial of degree d with roots $\alpha_1, \ldots, \alpha_d$.

Theorem 1.1 (Pandrosion identities for Δ_K). *All identities from Papers 14–30 apply verbatim:*

$$\Delta'_K(t) = \sum_{k=1}^{d} Q(\alpha_k, t), \tag{1}$$

$$E_{\Delta_K}(t) = \Delta'_K(t)^2 - \Delta_K(t)\,\Delta''_K(t) = \sum_{k=1}^{d} Q(\alpha_k, t)^2, \tag{2}$$

$$\zeta_K(1) := \sum_{k=1}^{d} \frac{1}{\Delta'_K(\alpha_k)} = 0, \tag{3}$$

$$D(\Delta_K)^2 = (-1)^{d(d-1)/2} \prod_{k=1}^{d} \Delta'_K(\alpha_k). \tag{4}$$

2 The knot zeta function

Definition 2.1. The **Pandrosion knot zeta function** is:

$$\boxed{\zeta_K(s) := \sum_{k=1}^{d} \Delta'_K(\alpha_k)^{-s}.} \tag{5}$$

By Paper 30: $\zeta_K(1) = 0$ (vanishing identity). This holds for all knots with simple Alexander roots.

Knot	$\zeta_K(1)$	$\zeta_K(2)$	$\prod \Delta'(\alpha_k)$
3_1 (trefoil)	0	$-2/3$	3
4_1 (figure-eight)	0	$2/5$	-5
5_1	0	0	125
5_2	0	-0.054	939.25
6_1–6_3	0	various	various

3 Palindromic symmetry and root pairing

Proposition 3.1 (Root pairing)**.** *The Alexander polynomial satisfies $\Delta_K(t) = \pm\, t^{-d}\, \Delta_K(t^{-1})$. Therefore, if α is a root, so is $1/\alpha$. The Pandrosion quotients come in **conjugate pairs**:*

$$Q(\alpha, t) \quad \textit{and} \quad Q(1/\alpha, t).$$

For roots on the unit circle ($|\alpha| = 1$): $1/\alpha = \bar{\alpha}$, so the pair sum $Q(\alpha, t) + Q(\bar{\alpha}, t)$ has real coefficients.

4 Roots on the unit circle: fibered knots

Knot	All $\lvert\alpha_k\rvert = 1$?	Fibered?
3_1	✓	Yes
4_1	$\times$ ($\lvert r\rvert \approx 2.618, 0.382$)	Yes
5_1	✓	Yes
5_2	✓	Yes
6_1	$\times$	No
6_2, 6_3	$\times$	No

Remark 4.1. Not all fibered knots have all Alexander roots on the unit circle (e.g., 4_1), but all knots with that property are fibered. The Pandrosion energy at unit-circle roots satisfies $E_{\Delta_K}(\alpha_k) = \Delta'(\alpha_k)^2 \in \mathbb{R}_{>0}$, since $\Delta'(\alpha_k) \in i\mathbb{R}$ for conjugate-paired roots on the circle.

5 Mahler measure and hyperbolic volume

Proposition 5.1 (Per-root Mahler decomposition, Paper 27)**.**

$$m(\Delta_K) = \log|a_d| + \sum_{k=1}^{d} \log^+ |\alpha_k|. \tag{6}$$

For knots with all $|\alpha_k| = 1$: $m(\Delta_K) = \log|a_d|$. For monic Alexander polynomials: $m = 0$.

Remark 5.2 (Boyd conjecture). Boyd (2002) conjectured that for hyperbolic knots K: $\mathrm{Vol}(S^3 \setminus K) \approx c \cdot m(\Delta_K)$, where the constant c depends on the knot. The Pandrosion per-root decomposition gives $m = \sum \log^+ |\alpha_k|$, expressing the volume as a sum of per-root contributions. Roots on the unit circle contribute nothing; only roots "off-circle" ($|\alpha| > 1$) generate volume.

Knot	$m(\Delta_K)$	Roots on $\|t\| = 1$?
3_1	0.000	All
4_1	0.962	No
5_1	0.000	All
5_2	0.693	All
6_1	0.767	No
6_2, 6_3	1.087	No

6 The knot energy polynomial

Definition 6.1. The **Pandrosion knot energy** is the polynomial:

$$E_K(t) := \Delta'_K(t)^2 - \Delta_K(t)\,\Delta''_K(t) = \sum_{k=1}^{d} Q(\alpha_k, t)^2 \in \mathbb{Z}[t]. \tag{7}$$

The knot energy at $t = -1$ is related to the knot determinant: $\Delta(-1)^2 = \det(K)^2$ and $E_K(-1) = \Delta'(-1)^2 - \det(K) \cdot \Delta''(-1)$.

For the trefoil: $E_K(-1) = 3$, $\det(K) = 3$.

7 Conclusion

Knot theory	**Pandrosion**
Alexander polynomial Δ_K	Polynomial with Pandrosion structure
Knot determinant $\det(K)$	$\|\Delta_K(-1)\|$
Pandrosion discriminant	$D^2 = \prod \Delta'(\alpha_k)$
Knot zeta function	$\zeta_K(s) = \sum \Delta'(\alpha_k)^{-s},\ \ \zeta_K(1) = 0$
Root pairing $(\alpha, 1/\alpha)$	Conjugate Q-quotient pairs
Mahler measure	$\sum \log^+ \|\alpha_k\| \approx c \cdot \mathrm{Vol}(S^3 \setminus K)$
Knot energy	$E_K = \sum Q_k^2 \in \mathbb{Z}[t]$

The Alexander polynomial, viewed through the Pandrosion lens, carries richer structure than its classical formulation suggests: a vanishing knot zeta function, a per-root volume decomposition, and an integral energy polynomial.

References

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